

Assignment - 6.

$$1) S(t) = x_I(t) \cos 2\pi f_c t - x_Q(t) \sin 2\pi f_c t.$$

$$\tilde{x}(t) = x_I(t) + j x_Q(t).$$

Review: I-branch: $\cos 2\pi f_c t$.

Q branch: $-\sin(2\pi f_c t + \epsilon)$.

$$a) y_I(t) = \text{LPF}\{S(t) \cos(2\pi f_c t)\} = \frac{1}{2} x_I(t).$$

$$y_Q(t) = \text{LPF}\{-g S(t) \sin(2\pi f_c t + \epsilon)\}.$$

$$= \frac{g}{2} [x_Q(t) \cos \epsilon + x_I(t) \sin \epsilon].$$

$$b). y(t) = y_I(t) + j y_Q(t).$$

$$y(t) = \frac{1}{2} x_I(t) + j \frac{g}{2} [x_Q(t) \cos \epsilon + x_I(t) \sin \epsilon].$$

$$c) \tilde{x}(t) = x_I(t) + j x_Q(t).$$

$$\tilde{x}^*(t) = x_I(t) - j x_Q(t).$$

$$y(t) = \frac{1}{2} x_I(t) + j \frac{g}{2} [x_Q(t) \cos \epsilon + x_I(t) \sin \epsilon].$$

$$= x_I(t) \left(\frac{1}{2} + j \frac{g \sin \epsilon}{2} \right) + \left(j \frac{g \cos \epsilon}{2} \right) x_Q(t).$$

$$x_I(f) = \frac{x(f) + x^*(f)}{2}$$

$$x_Q(f) = \frac{x(f) - x^*(f)}{2j}$$

$$y(t) = \left(\frac{1 + jg \sin \epsilon + jg \cos \epsilon}{4} \right) x(t) +$$

$$\left(\frac{1 + jg \sin \epsilon - g \cos \epsilon}{4} \right) x^*(t)$$

$$d) \alpha = \frac{1}{2} (1 + g e^{j\epsilon}) \quad \beta = \frac{1}{2} (1 - g e^{j\epsilon})$$

$$c) \text{ Ideal case } g = 1, \epsilon = 0 \quad \alpha = 1, \beta = 0$$

$$y(t) = x(t). \quad \text{No IQ imbalance.}$$

$$e) y[n] = \alpha x[n] + \beta x^*[n]$$

$$g) \beta = 0 \quad y[n] = \alpha x[n]$$

$$\alpha = 1 + ja \quad y[n] = x[n] e^{ja}$$

Constellation rotates by angle a .

b) Effect of non-zero β .

$$y[n] = \alpha x[n] + \beta x^*[n].$$

Mirror image component added constellation becomes distorted, asymmetrical & elliptical

$$3. \quad y[n] = \alpha x[n] + \beta x^*[n] + w[n].$$

$$a) \quad \hat{x}[n] = (y[n] = \alpha x[n] + \beta x^*[n]).$$

$\rightarrow x^*[n]$ cannot be eliminated

$\rightarrow$ standard & linear equalizer fails

$\rightarrow$ widely linear equalizer.

$$\hat{x}[n] = G_1 y[n] + G_2 y^*[n].$$

Substitute $y^*[n] = \alpha^* x^*[n] + \beta^* x[n]$.

$$\hat{x}[n] = (G_1 \alpha + G_2 \beta^*) x[n] + (G_1 \beta + G_2 \alpha^*) x^*[n].$$

for perfect recovery: $G_1 \alpha + G_2 \beta^* = 1$

$$G_1 \beta + G_2 \alpha^* = 0.$$

$$G_1 = \frac{\alpha^*}{|\alpha|^2 - |\beta|^2}$$

$$G_2 = \frac{-\beta}{|\alpha|^2 - |\beta|^2}.$$

1) For pilot based estimation $\rightarrow$ ① transmit known symbols $x[n]$

② observe $y[n]$.

③ solve $y[n] = \alpha x[n] + \beta x^*[n]$.

L.S: $(y) = [x \ x^*]^T$
 $\downarrow$
 least sq.

4 OFDM with IQ imbalance

$$y(k) = \alpha A(k) x(k) + \rho H^*[-k] x^*[-k] + w(k)$$

5) Vector form. $y = \begin{bmatrix} y(k_1) \\ y(k_2) \\ \vdots \end{bmatrix}$

$$A = \begin{bmatrix} H[k_1] H[k_2] & H^*[-k_1] x^*[-k_1] \\ \vdots & \vdots \end{bmatrix} \quad \theta = \begin{bmatrix} \alpha \\ \rho \end{bmatrix}$$

$$y = A\theta + w.$$

6) LS estimate: $\hat{\theta} = (A^H A)^{-1} A^H y.$

$$\begin{bmatrix} \hat{\alpha} \\ \hat{\rho} \end{bmatrix} = (A^H A)^{-1} A^H y.$$

5) a) Basis function $\phi_1(t) = 1$ for $0 \leq t \leq 1$, 0 otherwise.

$$\phi_2(t) = 1 \text{ for } 1 \leq t < 2, 0 \text{ otherwise.}$$

$$\phi_3(t) = 1 \text{ for } 2 \leq t < 3, 0 \text{ otherwise.}$$

$$\phi_4(t) = 1 \text{ for } 3 \leq t < 4, 0 \text{ otherwise.}$$

b) Signal vector.

$$s_1 = [2, 1, -1, -1] \quad s_2 = [1, 1, 1, 1]$$

$$s_3 = [-2, 1, 1, 0] \quad s_4 = [1, -2, -2, 2]$$

c) Euclidean Distance

$$d_{ij} = \sqrt{\sum_{k=1}^N (s_i(k) - s_j(k))^2}$$

$$d_{12} = \sqrt{5} \quad d_{13} = \sqrt{29} \quad d_{14} = \sqrt{19} \quad d_{23} = \sqrt{14} \quad d_{24} = \sqrt{10} \\ d_{34} = \sqrt{26}$$

6. Bandwidth constraint given

$$R_b = 12 \text{ Mbps} \quad B = 3 \text{ MHz}$$

using Nyquist. $R_s \leq 2B = 6 \text{ M symbols/sec}$

a) minimum constellation size

$$R_b = R_s \log_2 M \quad 12 \times 10^6 \leq 6 \times 10^6 \log_2 M$$

$$\log_2 M \geq 2 \quad M \geq 4$$

$M = 4$ for M-PSK, M-QAM, M-PAM.

b) symbol rate.

$$R_s = \frac{R_b}{\log_2 M} = \frac{12}{2} = 6 \text{ M symbols/sec}$$

c) minimum distance expression.

$$M\text{-PAM} \quad d_{\min} = \sqrt{\frac{12E_s}{M^2-1}}$$

$$M\text{-PSK} \quad d_{\min} = 2\sqrt{E_s} \sin\left(\frac{\pi}{M}\right)$$

$$M\text{-QAM} \quad d_{\min} = \sqrt{\frac{6E_s}{M-1}}$$

7) QPSK constellation.

a) minimum distance Square: distance b/w (A, A) & $(-A, A)$
 $d = 2A$.

diamond: distance b/w $(\sqrt{2}A, 0)$ & $(0, \sqrt{2}A)$.

$$d = \sqrt{2A^2 + 2A^2} = 2A$$

verified: $d_{\min} = 2A$.

b) Average Energy.

$$\text{Square: } E = A^2 + A^2 = 2A^2$$

$$\text{diamond: } E = (\sqrt{2}A)^2 = 2A^2$$

c) Power efficiency: Both have $E_s = 2A^2$ $d_{\min} = 2A$

Conclusion: both constellations are equally power efficient

8 a) M-FSK signals are orthogonal, better noise immunity, less interference b/w signals
 → lower probability of error at low SNR

b) M-FSK uses fixed bandwidth.

M-FSK ~~band uses fixed~~ bandwidth increases with m .

9). $r_k = s_k e^{j2\pi\Delta f_k t}$

rotation per symbol $\alpha = 2\pi\Delta f T$

5) Total Rotation.

Given: $\Delta f = 100 \text{ Hz}$ $T = 1 \text{ ms}$

$$\alpha = 2\pi(100)(0.001) = 0.2\pi \text{ rad}$$

After 10 symbols: $\alpha_{\text{total}} = 10 \times 0.2\pi = 2\pi \text{ rad}$

10) $r_k = s_k e^{j2\pi\Delta f_k t} + n_k$

$$s_k = A e^{j\theta_k}$$

$$r_k = A e^{j\theta_k} e^{j2\pi\Delta f_k t} + n_k$$

$\phi(k) = 2\pi \Delta f_k T \rightarrow$ represents phase rotations

that increases linearly with symbol index since it depends on $k \rightarrow$ time varying.

$$S_k = I_k + j Q_k$$

$$Y_k = (I_k + j Q_k) e^{j\phi(k)}$$

$$\rightarrow [I_k \cos(2\pi \Delta f_k T) - Q_k \sin(2\pi \Delta f_k T)] +$$

$\times$ Inphase.

$$j [I_k \sin(2\pi \Delta f_k T) + Q_k \cos(2\pi \Delta f_k T)] \rightarrow \text{Quadrature.}$$

$$b) 2\pi \Delta f T \ll 1$$

$$k. 2\pi \Delta f T \ll 1 \quad \& \text{ assume } \cos \theta \sim 1 \\ \sin \theta \approx \theta.$$

$$Y_k \approx (I_k + j Q_k) (1 + j 2\pi \Delta f_k T) + n_k$$

$$Y_k \approx (I_k + j Q_k) + j (I_k + j Q_k) (2\pi \Delta f_k T) + n_k$$

distortion $\approx j S_k (2\pi \Delta f_k T)$

$$I_m \approx I_k - Q_k (2\pi \Delta f_k T)$$

$$Q_m \approx Q_k + I_k (2\pi \Delta f_k T)$$

$$11. R_b = 9600 \text{ bits/s}$$

for binary PAM $R_s = R_b$

$$B_{\min} = R_s/2 \Rightarrow B_{\min} = \frac{9600}{2} = 4800 \text{ Hz}$$

if No, since $B_{\min} = 4800 \text{ Hz}$.

$$B_{\text{available}} = 6000 \text{ Hz}$$

since $B_{\text{available}} < B_{\min}$. $\rightarrow$ physically impossible

$$12. Q_{AM} \Rightarrow k = \frac{R}{R_s} = \frac{9600 \text{ bits/sec}}{2400 \text{ sym/sec}} = 4 \text{ bits/sym}$$

$$M = 2^4 = 16 \text{ QAM}$$

$$f_c = \frac{300 + 3300}{2} = 1800 \text{ Hz}$$

$$\text{Roll off factor } (\alpha) \Rightarrow B = R_s(1 + \alpha)$$

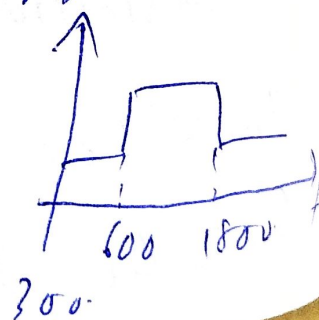
$$\text{available BW} \Rightarrow 3300 - 300 = 3000 \text{ Hz}$$

$$B = 3000 = 2400(1 + \alpha) \quad f_c = 1800 \text{ Hz}$$

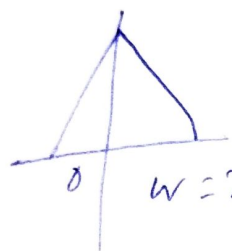
$$\text{Lower edge} = 1800 - 1500 = 300 \text{ Hz}$$

$$\text{upper edge} = 1800 + 1500 = 3300 \text{ Hz}$$

$$\text{required BW} \Rightarrow 1800 \pm 1200 = 600 \text{ to } 3000 \text{ Hz}$$



13) a) pulse rate for Nyquist criterion.



$$\omega_c = \frac{2\pi \times 10^6}{2} = \pi \times 10^6 \text{ rad/s.}$$

$$\omega_c = \frac{\pi}{T} = \pi \times 10^6 \Rightarrow T = 10^{-6}.$$

$$\text{pulse rate} = R_s = \frac{1}{T} = 10^6 \text{ pulses/sec.}$$

$$b) p(t) = F^{-1}(p(\omega)) = A \text{sinc}^2 B(t)$$

$$\text{peak } 10^{-6} \text{ \& } \omega = 2\pi \times 10^6 \quad p(t) = \text{sinc}^2(10^6 t)$$

$$\text{At } t = nT = n \times 10^{-6} \quad p(t) = 0 \quad \forall n \neq 0.$$

it satisfies Nyquist criterion

$$\text{transmission rate} \Rightarrow R_s \log_2 x = 10^6 \times 1 = 1 \text{ Mbps.}$$

c) Roll off.

$$\text{Abs bandwidth } (\omega) = 2\pi \times 10^6 \text{ rad/s.}$$

$$\text{Nyquist BW } \omega_0 \Rightarrow \omega_c = \pi \times 10^6 \text{ rad/s.}$$

$$\omega = \omega_0 (1+\alpha) \Rightarrow 2\pi \times 10^6 = 10^6 \times \pi (1+\alpha) \quad \alpha = 1.$$

roll off factor = 1.

$\Rightarrow$ 100% excess bandwidth.